



COMP232

Cybersecurity

Revision notes



Topics to revise

Part 1: Identification and authentication, Security Architecture

- Passwords v. tokens v. biometrics, Multifactor authentication
- Security attacks, passive v. active attacks, brute-force attack

Part 2: Cryptography

- Symmetric Encryption (DES, AES)
- Modes for block ciphers (ECB, CBC, etc)
- Public Key encryption (RSA), Key exchange (DH)
- Message authentication and hash functions (collision properties, birthday attack)
- Quantum cryptography (BB '84, QRNG)

Topics to revise

Part 2: Cryptography (cont.)

- **Advanced crypto:**
 - Fully homomorphic encryption and CryptDB
 - Zero-knowledge proofs
 - Secure Multiparty Computations
 - Steganography

Part 3: Security Protocols and their analysis

- NS authentication protocol, man-in-the-middle attack, protection against it
- Formal methods for protocol analysis, model checking

Topics to revise

Part 4: Monitoring and intrusion detection

- Intrusion detection (anomaly and signature based)
- IDS Techniques
- Firewalls (and their types)
- Attacks and defences
 - Viruses, worms, anti-virus and anti-worm methods

Part 5: Privacy and Anonymity, Ethical Issues

- Mix-networks, Crowds, Tor-network, etc
- Legal Aspects: Motivations, OECD Guidelines, etc

How to revise

Possible way:

Go through the list of topics (+ titles of individual slides of all lectures) and check if you can say something **precise** (a few sentences) in response to the questions related to the topic/slide title:

What? How? Why?